UAB
Universitat Autònoma de Barcelona

GROUP 10 - FINAL PRESENTATION

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ICD-10 CODIFICATION FNL PROJECT

asho

kaggle

1. CORE DIFFERENTIATORS

CONSERVATIVE PREPROCESSING

- Intentionally protected clinical noise to preserve subword tokenizer fidelity

SAFE DATA HANDLING

- Rejected destructive data augmentation to prevent corrupting medical negation and context

HYBRID ENSEMBLING

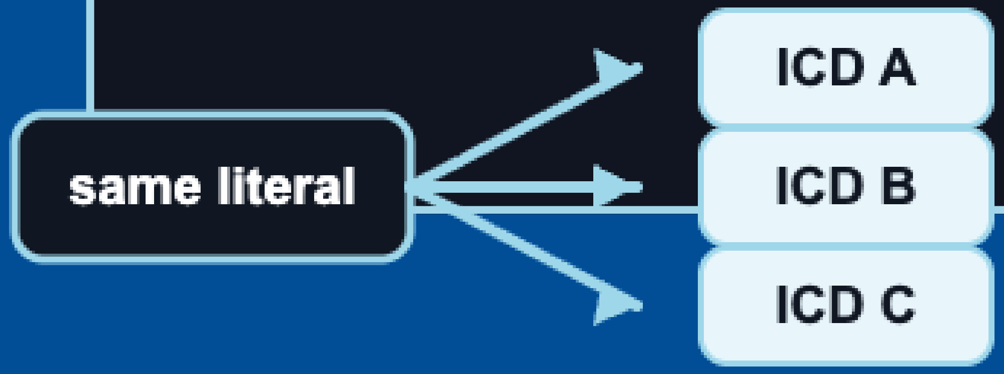
- Fused classical VSM (tf-idf) with contextual encoders (RoBERTa) to capture uncorrelated errors

2. EDA & MODELING DECISIONS

Literal Collisions

- We found ≈1 500 identical literal groups mapping to different ICD-10 codes.

- NO LOOKUP TABLE



Code Representation Imbalance

- Some codes are overrepresented (‘Z’, ‘O’)

- MOVED FROM ACC. TO MACRO-F1

CHARACTER PRESENCE CONSISTENCY: TEST - TRAIN

- Accent & Digit occurrances were consistent across both train and test.

- SUPPORTS KEEPING DATA ‘DIRTY’

3. PREPROCESSING

PIPELINE:

- String verification, whitespace normalization via regex, leading space stripping

DISCOVERY:

- Aggressive normalization severely damaged downstram model signal

WHY?

- Clinical texts are short (17 ave length). Stripping them destroys vocabulary alignment for pretrained models

4. ML APPROACH & BENCHMARKS

Majority Baseline

- By always predicting the MODE class:
- ≈ 0.12 acc (floor accuracy baseline)

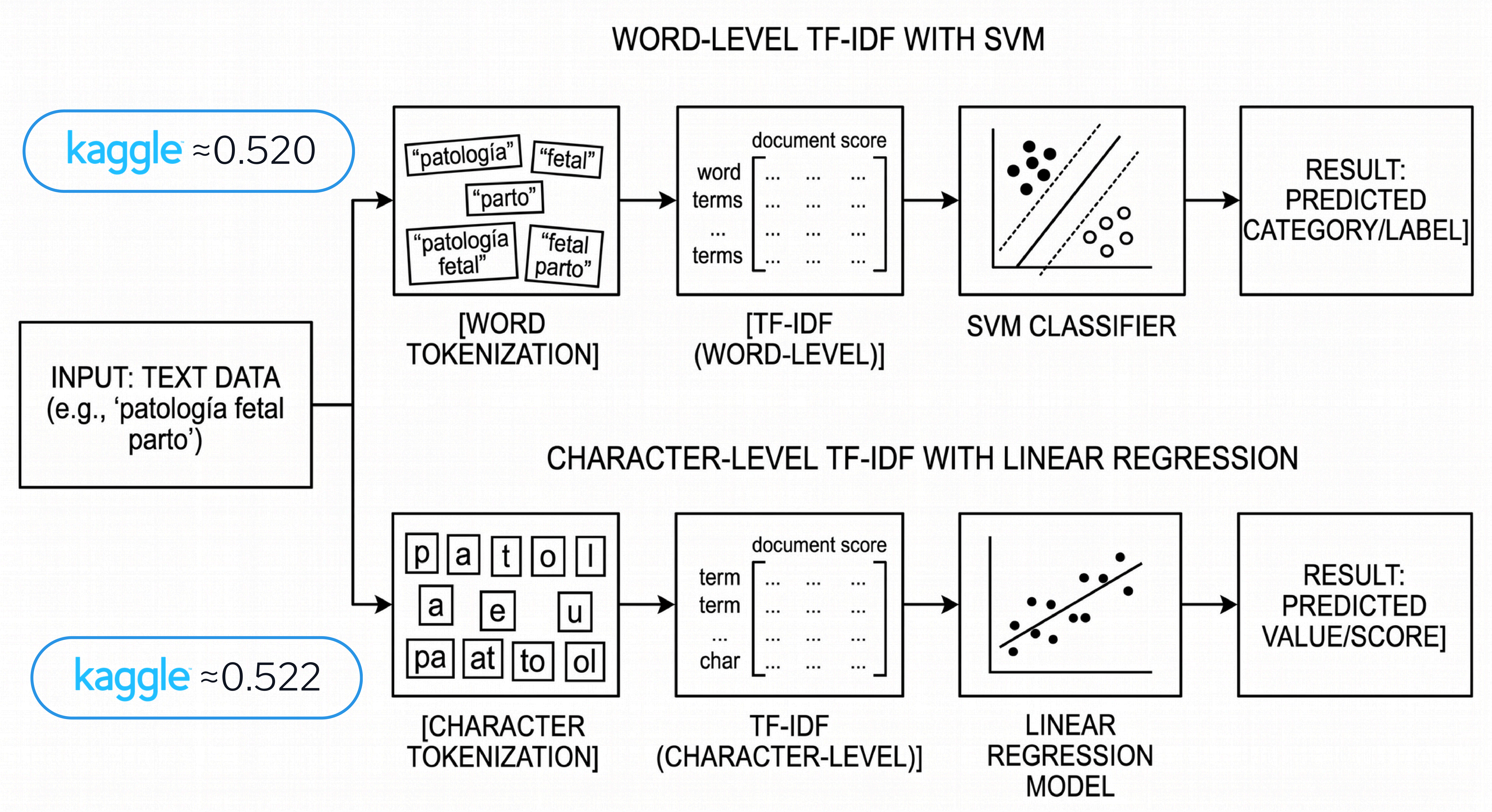
Character LevelTF-IDF + Regression

- ACC jumps to ≈ 0.52
- Macro-F1 ≈ 0.40

Word Level TF-IDF + Linear SVM

- Captures whole lexical units.
- ACC ≈ 0.52
- Macro-F1 ≈ 0.47
- BEST RESULTS!

4. ML APPROACH & BENCHMARKS



Let's try Deep Learning!

Hugging Face

PlanTL-GOB-ES/**roberta-base-biomedical-clinical-es**

- cross entrolpy loss

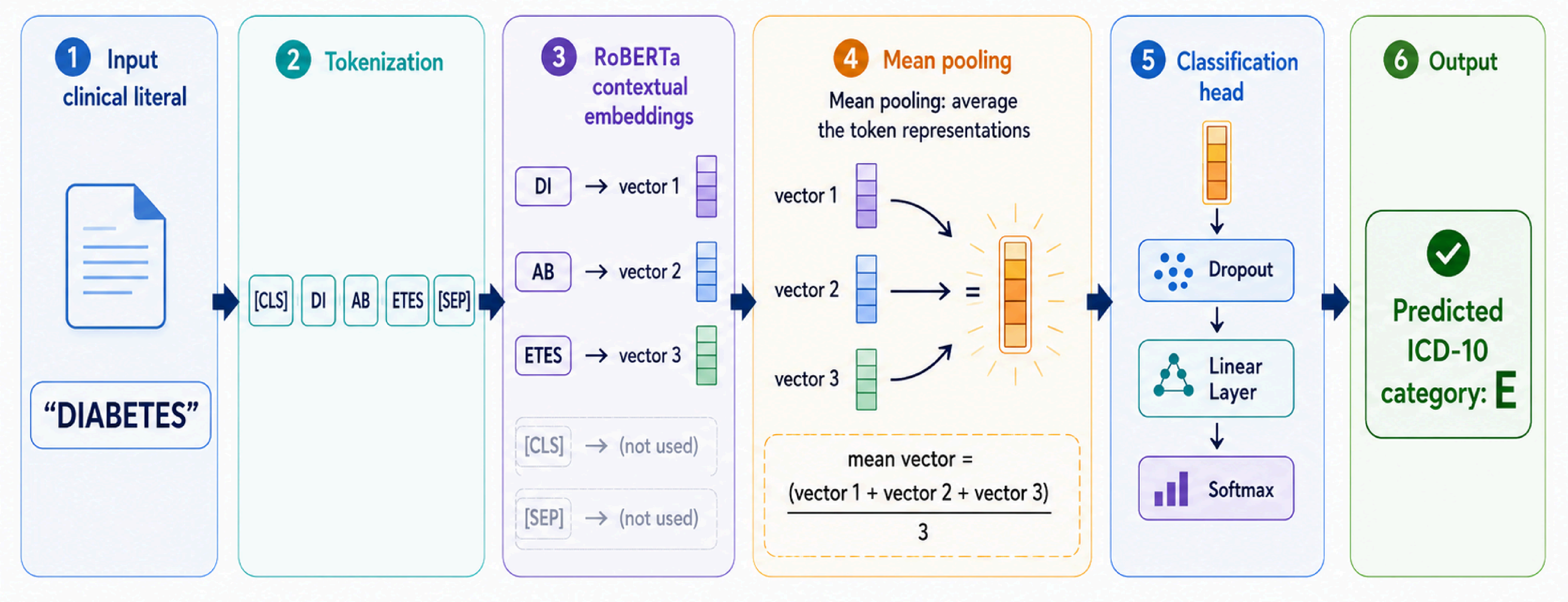
$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp z_{i,y_i}}{\sum_{c=1}^{C} \exp z_{i,c}},$$

- early stopping

Hyperparameters	Value
Max epochs	50
Patience	10
Batch size	128
Learning rate	2e-5
Weight decay	0.01
Dropout	0.1

Fine-tuning with Mean Pooling

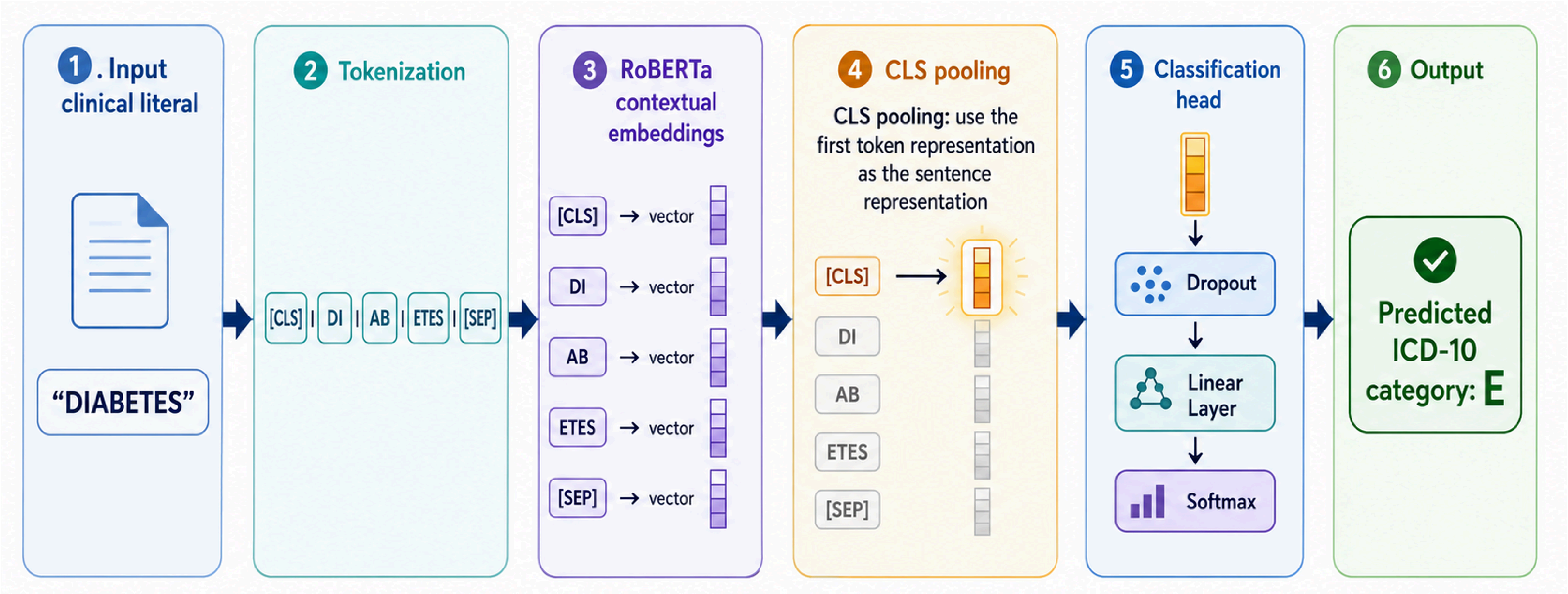
Macro-F1≈0.494
F1-weighted≈0.554



kaggle ≈0.564

Fine-tuning with CLS

Macro-F1≈0.496
F1-weighted≈0.550



kaggle ≈0.569

TF-IDF

Strenghts

- caputers textual clues + lexical baseline

  → great for short literals

  → ex. punctuation, digits, accents...

Weaknesses

- no biomedical meaning
- generalizes synonyms + unseen words

  → similar words with different categories

RoBERTa

Strenghts

- maps context + Token relationships

  → BIOMEDICAL SEMANTIC MEANING

Weaknesses

- needs contexts

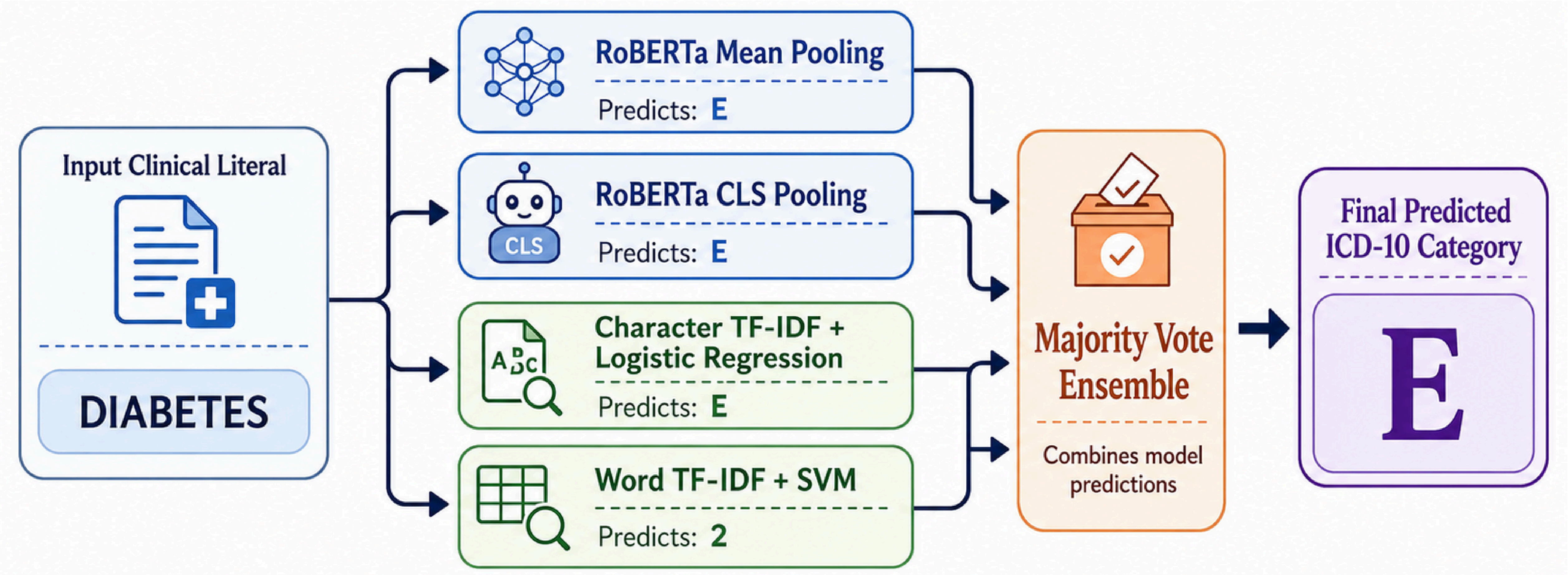
  → our literals are too short

  → mean train literal length ≈ 16.95 chars

- affected by contradictory labels

  → 1,486 duplicate literal groups with multiple categories

Let's apply Machine Learning!



Macro-F1≈0.496
F1-weighted≈0.561

ACC

how often the system gives the correct code overall

F1

whether the system is reliable for rare categories

W-F1

whether the system is reliable for common categories

Version	Model	Accuracy	Macro-F1	Weighted-F1	Public
v00	Majority baseline	0.125182	0.006181	0.027854	–
v01	Char TF–IDF + logreg	0.522628	0.402554	0.494943	0.550
v02	Word TF–IDF + SVM	0.520073	0.474196	0.514018	0.536
v03	Similarity retrieval	0.497445	0.462789	0.496120	–
v04	RoBERTa CLS	0.569343	0.494329	0.554347	0.573
v05	RoBERTa mean	0.564599	0.496567	0.549541	0.556
v06	Imbalance-aware RoBERTa	0.557299	0.480394	0.539776	0.555
v07	Tuned RoBERTa mean	0.564599	0.494151	0.549054	0.572
v08	Safe data strategy	0.568613	0.481648	0.549150	0.583
v09	Validation ensemble	0.576642	**0.506277**	**0.561544**	0.573
v10	Diverse ensemble	**0.579562**	0.496677	0.561294	**0.587**

BONUS Leaderboard Evaluation

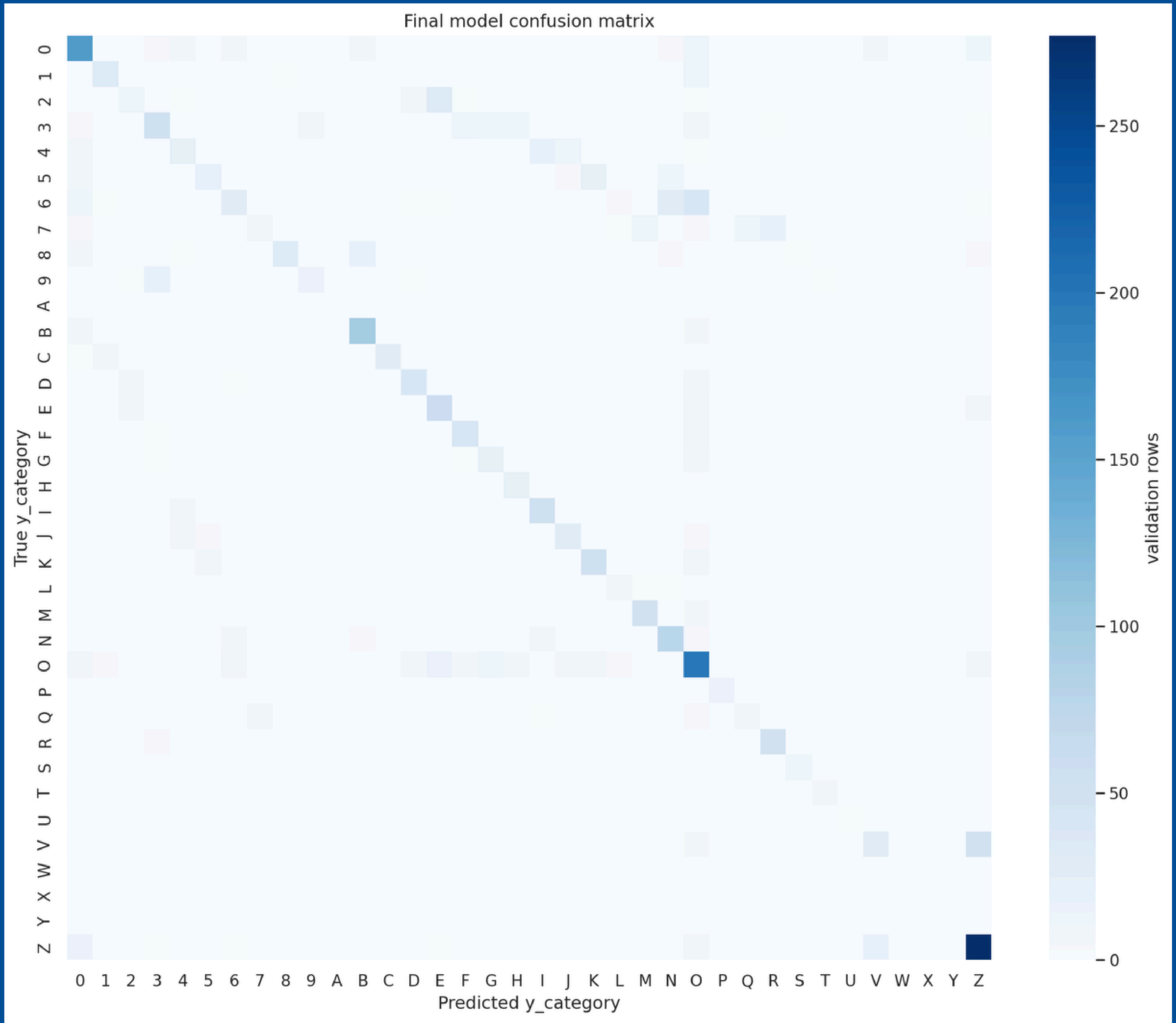
#	△	Team	Members	Score	Entries	Last	Solution
1	—	Group 20		0.598	81	13h	
2	▲ 1	Team 3		0.588	18	5d	
3	▼ 1	group 4		0.587	53	4d	
4	▲ 1	Group 2		0.577	7	21d	
5	▲ 5	Joan A		0.571	5	12h	
6	▼ 2	Group 10		0.569	20		

9. ERROR ANALYSIS

CONFUSION MATRIX

Confusion matrix reveals medicine-category overlaps:

- True V with Pred Z
- True 6 with Pred O



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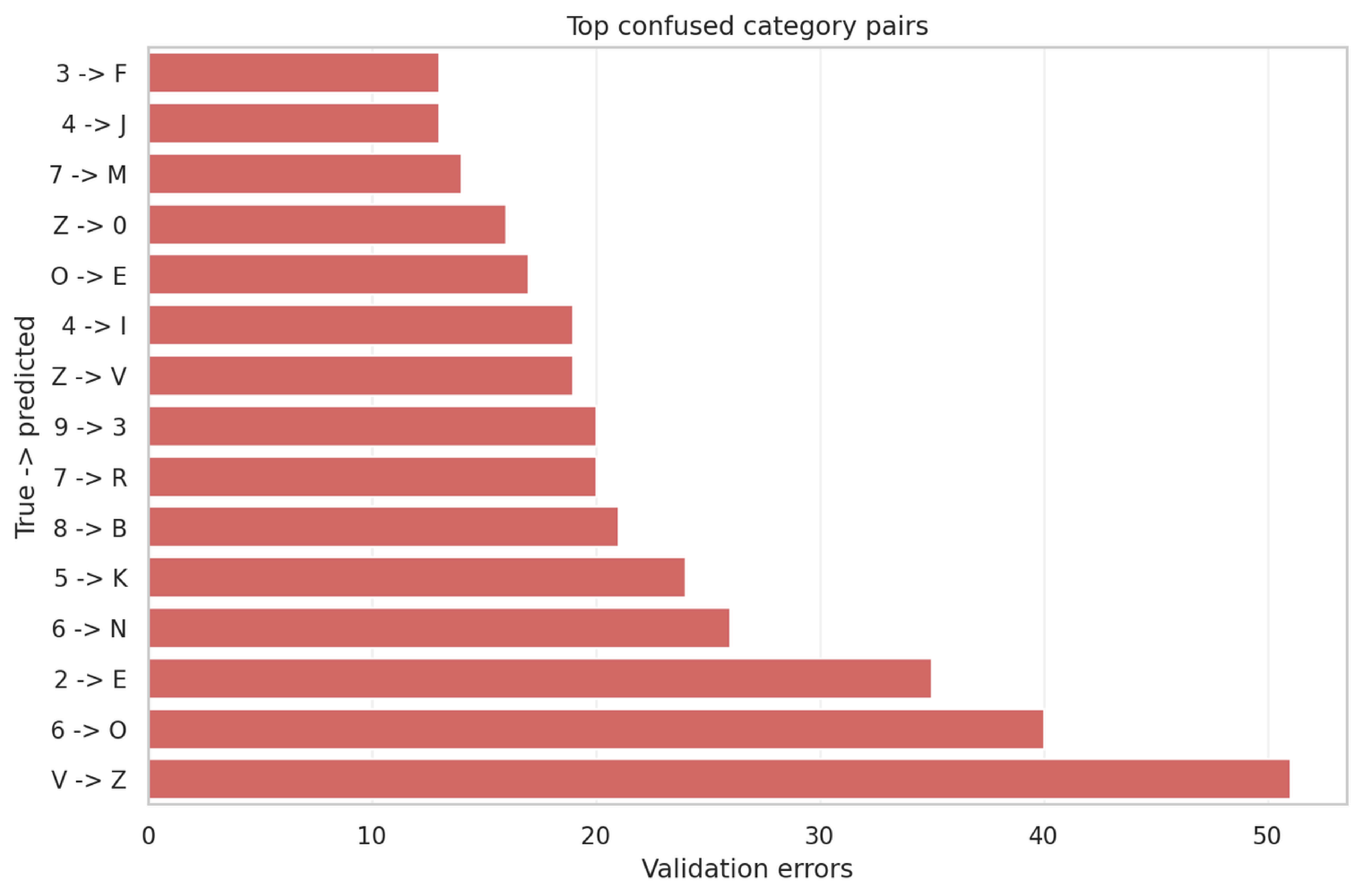

10. CLINICAL REALITY AND LIMITATIONS

THE MODEL IS NOT PERFECT

- BEST SCORE of 0.587 demonstrates the model **learnt** patterns and information hidden in the literals.
- The model is NOT READY for UNSUPERVISED deployment in a critical sector like the medical.

USEFUL AS A DECISION-SUPPORT TOOL

Thank you!

Do you have any questions?